

IS 3053 – Data Mining Techniques

Group Assignment

ANALYSIS ON EMPLOYEE DATASET



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1.0 Objective of the Analysis

The primary objective of this analysis is to identify homogeneous groups of employees based on job-related characteristics and to build predictive models for salary estimation. Specifically, this study aims to:

- **Cluster Employees:** Utilize unsupervised learning (K-Means clustering) to group employees based on variables such as Age, Educational Level, Employment Category, Salary, and Experience.
- **Analyze Features:** Interpret the common characteristics of the identified clusters to understand the distinct profiles within the workforce.
- **Predict Salary:** Develop separate regression tree models for male and female employees within each cluster to estimate current salary based on demographic and professional factors

2.0 Variable Description:

The analysis is based on the dataset employee_data.csv, which initially consisted of 474 records containing 10 variables. The dataset includes both demographic information (e.g., Gender, Date of Birth) and professional details (e.g., Salary, Job Category).

Column Name	Data Type	Description
id	Integer	Unique identifier for each employee.
gender	Categorical	Gender of the employee (Male/Female).
bdate	Date	Birth date of the employee.
educ	Integer	Total years of education completed.
jobcat	Categorical	Employment category (1: Clerical, 2: Custodial, 3: Manager).
salary	Integer	Current annual salary in dollars.
salbegin	Integer	Annual salary at the time of hiring.
jobtime	Integer	Number of months since the employee was hired.
prevexp	Integer	Previous work experience in months before joining.
minority	Categorical	Minority status indicator (0: Nonminority, 1: Minority).

3.0 Pre-processing of Data

Prior to analysis, the data underwent a rigorous pre-processing pipeline to ensure quality and compatibility with machine learning algorithms. **Data Cleaning**

- **Duplicate Handling:** The dataset was checked for duplicate records. 0 duplicate rows were identified.

- **Missing Values:** Consistent with the dataset documentation, specific codes were treated as missing values: 0 for variables such as Education and Salary, and 9 for Minority Classification. A total of 25 rows containing these codes or standard missing values were removed, resulting in a final dataset of 449 records. **Feature Engineering**
- **Age Calculation:** The original dataset provided bdate (Date of Birth). To utilize age as a clustering variable, a new variable Age was derived by calculating the difference between the birth date and a reference date of January 1, 1995.

One-Hot Encoding

The variable jobcat represents categorical data (1=Clerical, 2=Custodial, 3=Manager). To avoid the clustering algorithm incorrectly interpreting these categories as an ordinal sequence (where "Manager" is mathematically 3 times "Clerical"), **One-Hot Encoding** was applied. This process transformed the single jobcat column into three binary variables (job_Clerical, job_Custodial, job_Manager), allowing the algorithm to treat each category as a distinct entity.

Data Splitting To evaluate the predictive performance of the regression trees, the data was split into training and testing sets. A random split of **80% Training** (360 samples) and **20% Testing** (89 samples) was employed.

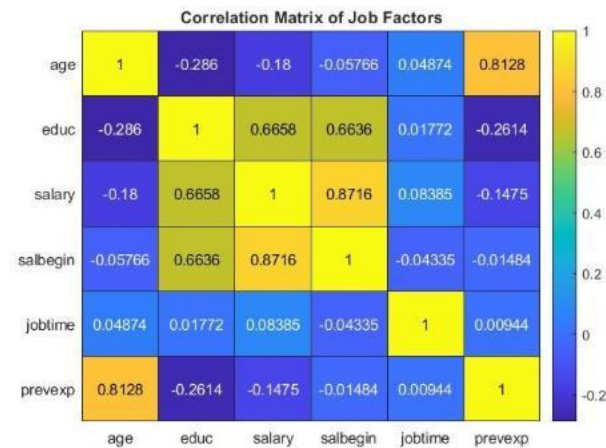
Data Scaling Since the K-Means clustering algorithm relies on Euclidean distance, it is sensitive to the scale of variables. The dataset contains variables with vastly different magnitudes (e.g., salary in tens of thousands versus educ in years). To address this, **Z-Score Standardization** was applied.

- The mean and standard deviation were calculated using **only the Training set** to prevent data leakage.
- These parameters were then used to scale both the Training and Test sets, ensuring all variables have a mean of 0 and a standard deviation of 1.

4.0 Descriptive data analysis

Correlation Heatmap

Starting salary is the best predictor of current salary (\$0.87\$), meaning your initial pay strongly determines your future pay.

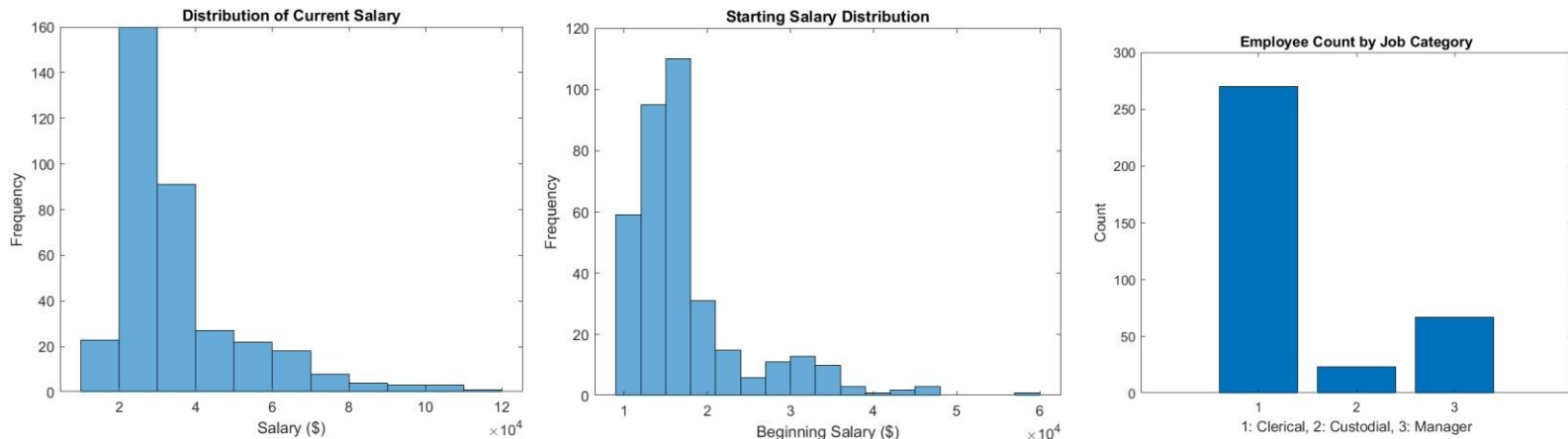


There is a strong positive link between education and salary (\$0.67\$); generally, more schooling leads to higher earnings.

Age and previous work experience are very highly correlated (\$0.81\$), which makes sense as older employees usually have longer work histories.

Factors like age (\$-0.18\$) and previous experience (\$-0.15\$) actually have slight negative correlations with salary in this specific dataset.

4.1 Univariate Analysis

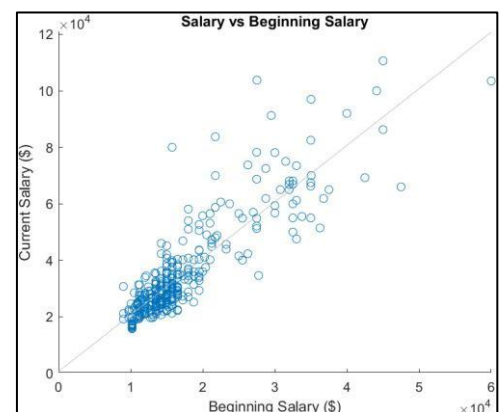


The current salary and starting salary exhibit positive skewness. According to job categories, The workforce is predominantly **clerical** compared to others.

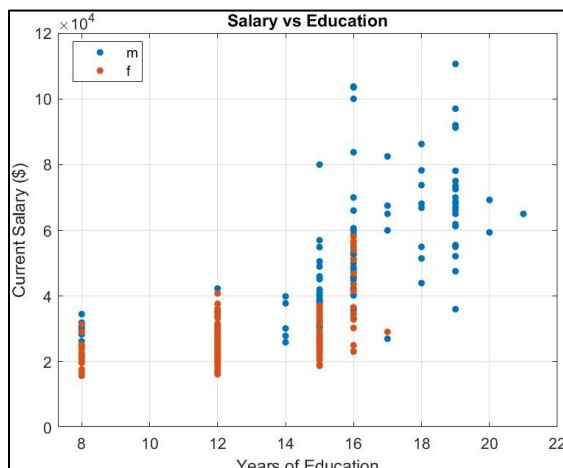
4.2 Bivariate Analysis

Salary vs Beginning Salary

There is a very strong positive correlation between an employee's starting salary and their current salary. The data points follow the trendline closely, meaning those who started with higher pay generally remain the highest earners over time. This suggests that the initial starting salary is one of the most reliable predictors of what they earn today.



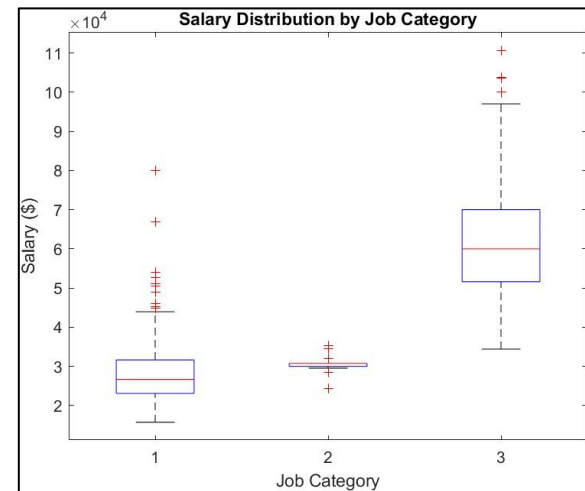
Education vs Salary



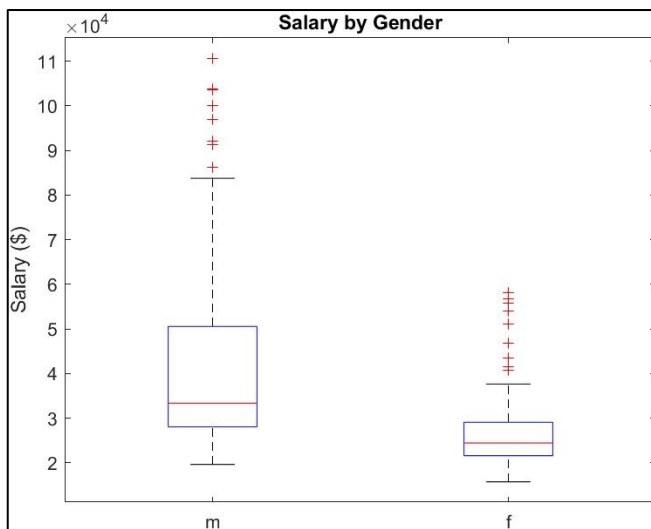
Salary generally increases with higher education, but a clear gender pay gap exists. Men consistently earn more than women across all education levels shown. The disparity is most extreme at higher education levels, where the top earners are all male. Women remain concentrated in lower-paying positions regardless of their years of schooling.

Salary vs Job Category

Managers earn the highest and most spread out salaries, reaching over \$110,000. Clerical and Custodial roles have much lower pay and tighter salary ranges. The Clerical group has several "outliers," which are specific people earning much more than the rest of their group. Overall, the data shows that job category is a very strong predictor of how much an employee earns.

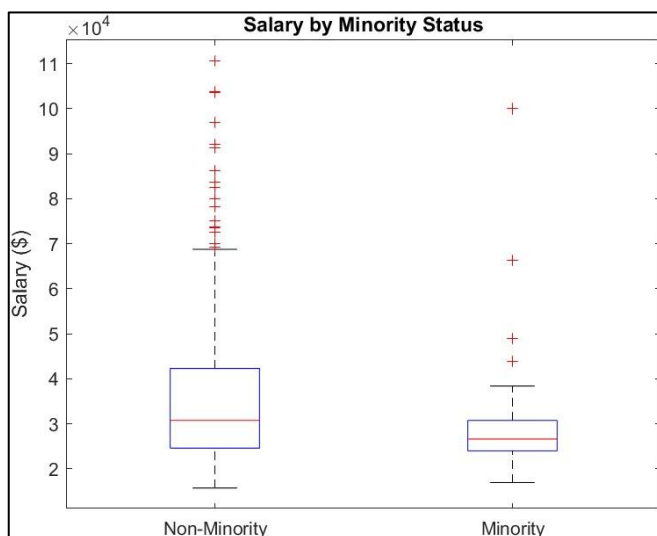


Salary vs Gender



Men consistently earn higher salaries than women across the entire organization. All top-tier salaries over \$60,000 are held by male employees, while female pay is mostly concentrated at the lower end. This pay gap remains even when men and women have the same level of education. The highest male earners reach over \$110,000, which is nearly double the highest female salary.

Salary vs Minority Status



Non-Minority employees have a higher median salary and a much wider pay range than Minority employees. The highest earners (reaching over \$110,000) are almost exclusively in the Non-Minority group. Minority employees are mostly concentrated in lower-paying roles with very few reaching high salary levels.

5.0 Advanced Data Analysis

5.1 Cluster Analysis

Dividing employees into similar groups helps identify common workforce patterns and understand structural differences within the organization. It also assists in recognizing

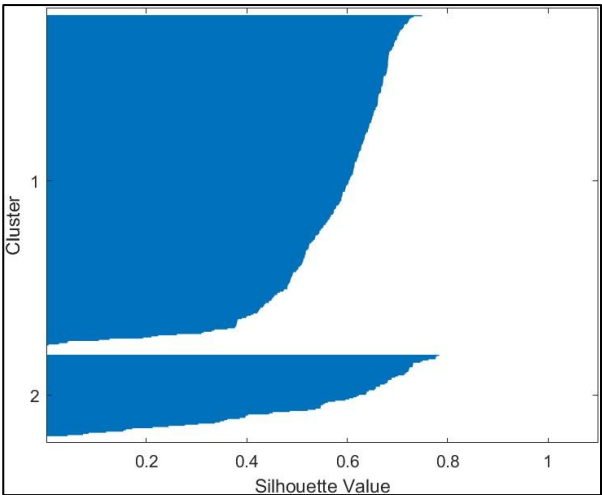
groups with distinct salary levels, experience profiles, and job categories. To achieve this, cluster analysis was performed to group employees based on their similarities in job-related characteristics.

Since the dataset contains categorical variables such as job Category and Minority Classification, these variables were first transformed using **one-hot encoding** to convert them into numerical format. After preprocessing and standardizing the numerical variables to remove scale effects, the **K-means clustering algorithm** was applied to the training data).

The clustering was performed using the **squared Euclidean distance** as the distance measure. To determine the optimal number of clusters, 2 to 6 no of clusters were tested and evaluated using the **average silhouette score**. The highest silhouette score obtained was **0.5638**, corresponding to **2 clusters**, indicating that this is the optimal cluster solution.

Cluster 1 contains more employees than Cluster 2, but both clusters demonstrate clear separation. Therefore, dividing the employees into two clusters is an appropriate and meaningful segmentation of the dataset.

No.of Clusters	Average Silhouette score
2	0.5638
3	0.5358
4	0.4126
5	0.4064
6	0.3818



Common features of the employees who belong to the identified clusters .

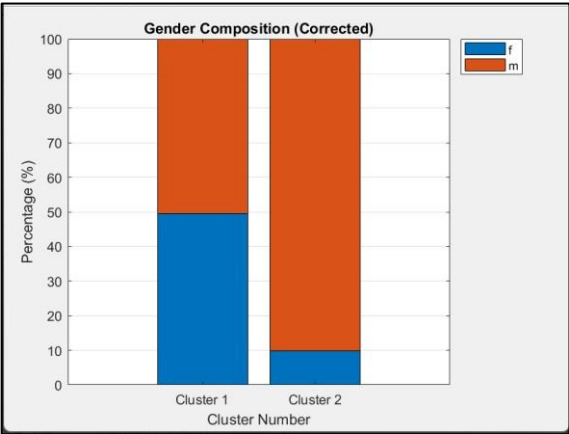


Figure : Percentage Distribution of Genders within Clusters

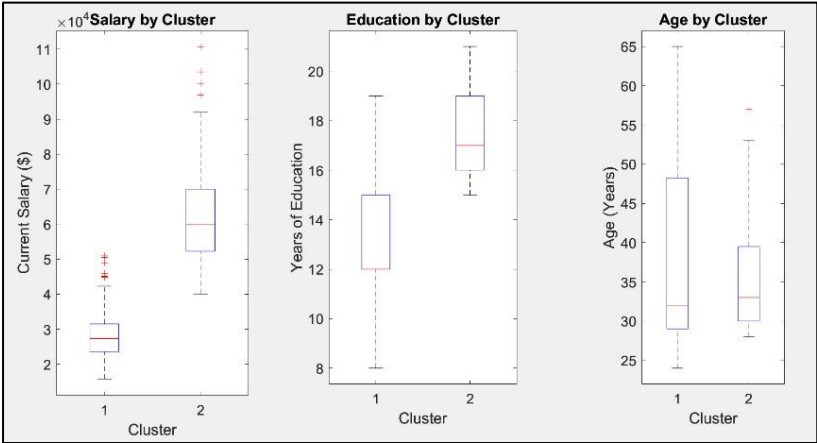


Figure : Comparison of Current Salary, Education Level, and Age across

Based on the visual analysis of the clustered data (Figures X, Y, and Z), the algorithm successfully partitioned the employees into two highly distinct profiles:

- **Cluster 1 (The General Workforce) Job & Pay:** Represents the majority of the workforce, consisting exclusively of Clerical and Custodial roles. Consequently, they fall into the lower salary bracket (median ~\$28,000).
 - **Demographics & Education:** This group has an average education level of 12 years (high school equivalent) and spans a wide age range. Demographically, it features an even, nearly 50/50 split between male and female employees.
- **Cluster 2 (The Managerial/Professional Workforce):**
 - **Job & Pay:** A smaller, elite group composed overwhelmingly of Managers. This group commands a significantly higher compensation (median ~\$60,000).
 - **Demographics & Education:** Employees here are highly educated, with a median of 17 years of education (bachelor's degree or higher). Demographically, this group is heavily skewed, consisting of roughly 90% male employees.

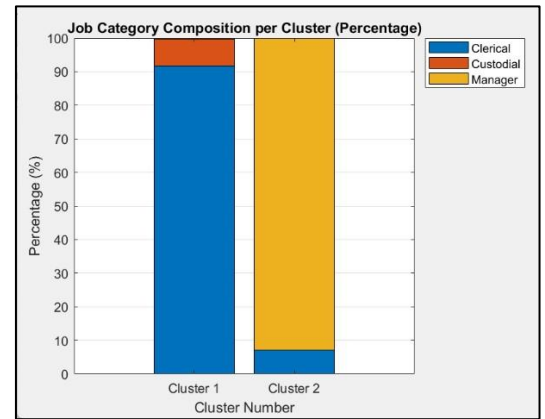


Figure : Distribution of Genders within clusters

5.2 Model Selection Strategy

The objective of this analysis was to develop **regression trees** to estimate the current salary of males and females separately.

To determine the optimal modeling strategy, an initial feasibility test was conducted by stratifying the data into four distinct subgroups based on Gender and Cluster assignment. The distribution of employees across these groups is summarized in Table 1 below:

Justification for Rejection of subgroup Approach: As shown in Table 1, The critical issue arises within the

Subgroup	Cluster	Count (n)	Cluster Distribution
Male	Cluster 1	146	80.28% (Total Cluster 1)
Female	Cluster 1	143	
Male	Cluster 2	64	19.72% (Total Cluster 2)
Female	Cluster 2	7	

Table 1: Sample Size Distribution by Gender and Cluster

"Female - Cluster 2" subgroup, which contains only **7 observations**. A sample size of $n=7$ is statistically inadequate for training a regression tree. When a model was attempted for this subgroup, the algorithm failed to identify significant splitting variables, resulting in a single-node model with no predictive logic.

Consequently, the subgroup approach was rejected to avoid overfitting and instability. Furthermore, the instability of this segmented approach was confirmed by the error metrics. As detailed in Table 2, the Root Mean Square Error (RMSE) fluctuated drastically between subgroups.

Model Subgroup	MSE (Mean Squared Error)	RMSE (Root Mean Squared Error)
Female – Cluster 1	2.29×10^7	\$4,786\$
Male – Cluster 1	3.53×10^7	\$5,942\$
Female – Cluster 2	3.74×10^7	\$6,116\$
Male – Cluster 2	3.38×10^8	\$18,371\$

The results in Table 2 highlight extreme variance in model performance. The RMSE for the "Male - Cluster 2" group \$18,371 is more than four times higher than that of the "Female - Cluster 1" group \$4,398. This inconsistency confirms that the subgroups partitions lacked the statistical density required for reliable prediction across all groups.

Final Approach:

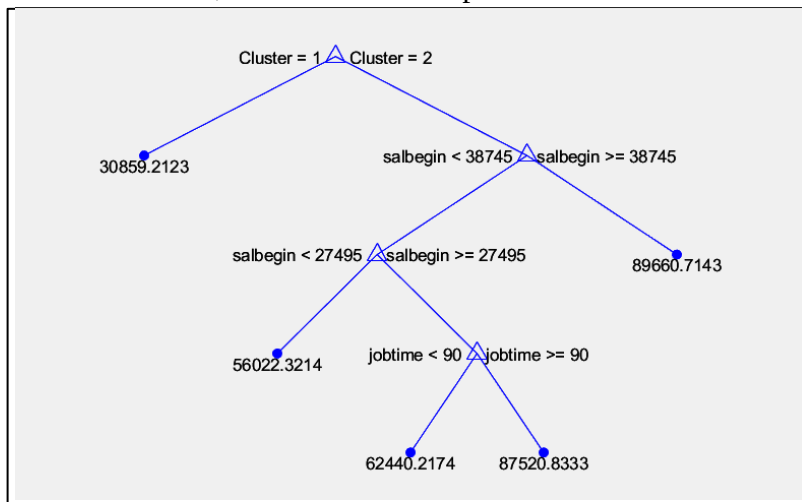
To ensure statistical robustness and prevent overfitting, the final analysis adopted a "Two-Model Approach":

- **Male Model:** Training a single regression tree for all male employees (**n=210**).
- **Female Model:** Training a single regression tree for all female employees (**n=150**).

To retain the value of our clustering, we added the **Cluster ID** as an input variable for the trees. This gives the models a simple way to separate managers from general staff without sacrificing the statistical power of our larger dataset.

5.2 Regression Tree Analysis: Male Employees

The regression tree for male employees utilized Age, Educational Level, Job Time, Previous Experience, Minority Classification, and Cluster ID as predictors to estimate Current Salary.

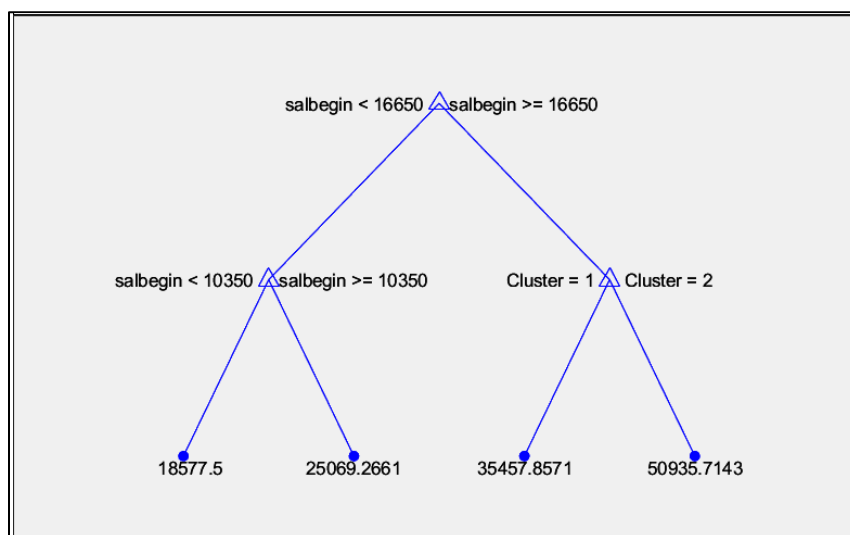


(Figure 1: Regression Tree for Male Employees - Root Node: Cluster)**Interpretation:** As shown in **Figure 1**, the algorithm identified **Cluster** as the root node (the most significant splitting variable).

employees based on their cluster assignment. This indicates that for male employees, their broad job profile is the single strongest determinant of their current salary band. • **Secondary Splits:** Within these clusters, the tree further refines predictions using salbegin (Beginning Salary) and jobtime (Months since

Hire). This suggests that once the job category (Cluster) is established, experience and tenure become the primary drivers of salary growth.

5.3 Regression Tree Analysis: Female Employees



A separate regression tree was developed for female employees using the same set of predictor variables.

(Figure 2: Regression Tree for Female Employees - Root Node: Beginning Salary)

Interpretation: As shown in Figure 2, the regression tree for females exhibits a different structure compared to the male model.

- **Primary Split:** The root node is Beginning Salary rather than Cluster.
- **Secondary Splits:** The Cluster variable appears as a secondary splitting

variable lower down the tree (on the right branch).

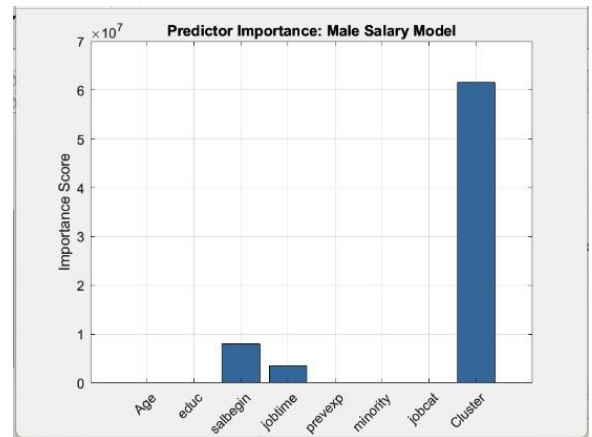
- Analysis: This structure suggests that for female employees, current salary is strictly tied to their entry level pay. While the Cluster grouping is statistically significant (as it appears in the tree), it is secondary to the raw financial starting point. This indicates a more rigid salary structure for females, where advancement is linearly tied to starting pay rather than broad job categories.

5.4 Feature Importance Analysis

- Predictor importance scores were analyzed to quantify which variables contributed most to reducing prediction error in each model.

A. Male Model Feature Importance

As shown in **Figure 3**, the **Cluster** variable is by far the most important factor for predicting male salaries. Its bar is massively taller than the rest. **Beginning Salary (salbegin)** and **Job Time (jobtime)** are the only other variables that matter, but their impact is very small in comparison. This tells us a clear story: for men, getting into a higher job category (like management) is the main driver of a high salary, while their starting pay matters much less



(Figure 3: Predictor Importance for Male Salary Mode)

• B. Female Model Feature Importance

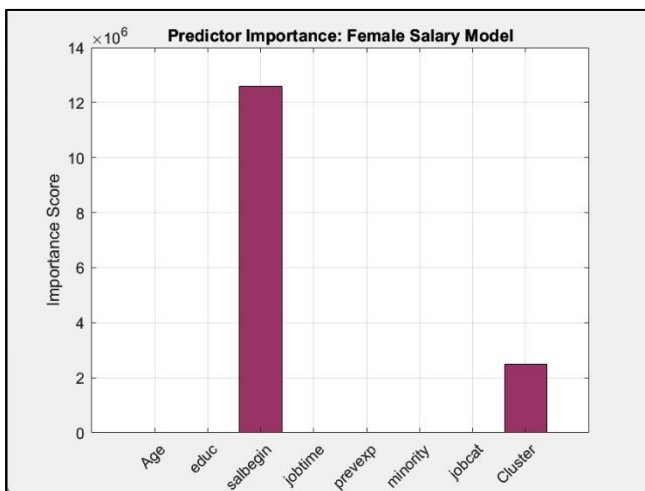


Figure 4 shows a completely different picture for female employees. **Starting Salary (salbegin)** is by far the most important factor, as shown by the massive bar on the chart. The job group (Cluster) matters only a little bit, and the other details do not matter at all. This means that for women, their current pay is mostly decided by what they were paid on their first day, rather than their job category.

(Figure 4: Predictor Importance for Female Salary Model)

5.5 Model Performance Evaluation

To assess the predictive accuracy of the final regression trees, the Root Mean Square Error (RMSE) was calculated for both models. RMSE provides a measure of the average deviation between the predicted salary and the actual salary in US Dollars.

Table 4: Performance Metrics for Method 2 (Two-Model Approach)

Model	MSE (Mean Squared Error)	RMSE (Root Mean Squared Error)	Performance Interpretation
Male Model	\$1.19 10 ⁸	\$10,922	Higher variance due to wider salary ranges in managerial roles.
Female Model	\$2.36 10 ⁷	\$4,858	Lower variance indicating a highly predictable, linear salary structure.

6.0 Conclusion:

As shown in Table 4, the Female Model achieved a significantly lower RMSE \$4,858 compared to the Male Model (\$10,922).

- **Female Accuracy:** The low error rate suggests that female salaries in this dataset follow a strict, predictable pattern driven largely by starting salary (salbegin), with very few outliers.
- **Male Variance:** The higher error rate for males reflects greater variability in the upper salary brackets. Since the male dataset contains more high-earning managers (Cluster 2), there is naturally more fluctuation in their compensation that simple demographic variables cannot fully capture.

Overall, the Two-Model Approach successfully stabilized the predictions, achieving robust accuracy for the majority of the workforce while highlighting the structural differences in pay between genders.

The analysis successfully combined clustering and regression trees to model employee salaries and understand workforce structure. Two distinct clusters were identified, separating general staff from managerial employees. The final two-model approach (separate models for males and females with cluster as a predictor) produced stable and reliable predictions. Results showed that male salaries are mainly influenced by job category, while female salaries are more strongly determined by starting salary. Overall, the approach achieved good predictive accuracy and revealed important structural differences in salary progression.